

What is Web Scraping?

Web scraping is the process of **automatically collecting data from websites** using a program instead of copying it manually.

In simple words:

👉 *A program visits a website, reads the page content, and extracts useful information.*

Example:

- Collecting product prices from an e-commerce site
- Extracting news headlines from a news website
- Gathering job listings from a job portal

Use Cases of Web Scraping

Web scraping is widely used in many fields:

Data Collection & Analysis

- Price comparison websites
- Market research
- Stock or crypto data analysis

Machine Learning & AI

- Collecting datasets for training models
- Text analysis (reviews, comments, articles)

Education & Research

- Academic research data
- Educational projects

Business & Automation

- Monitoring competitors' prices
- Tracking trends and reviews

Ethics of Web Scraping

Just because data is online **does not mean it should be scraped freely**.

✓ Ethical Practices

- Scrape **publicly available data only**
- Respect website rules (`robots.txt`)
- Do not overload servers (use delays)
- Give credit when required

✗ Unethical Practices

- Scraping private or paid data
- Stealing copyrighted content
- Collecting personal user information

Legal Considerations

Web scraping laws vary by country and website.

Important Points:

- Always check a website's **Terms & Conditions**
- Some websites **explicitly forbid scraping**
- Scraping personal data may violate **privacy laws**
- Commercial use of scraped data can cause legal issues

Rule of thumb:

If the data is sensitive, private, or restricted → **don't scrape it**

Static vs Dynamic Websites

♦ Static Websites

- Content is already in **HTML**
- Loads once and doesn't change automatically
- Easy to scrape

Example:

Blogs, simple product pages, documentation sites

✓ Tools used: `requests`, `BeautifulSoup`

♦ **Dynamic Websites**

- Content loads using **JavaScript**
- Data appears after page loads or user interaction
- Harder to scrape

Example:

Facebook, YouTube, some job portals

✓ Tools used: `Selenium`, `Playwright` (advanced)



Installing Python Libraries for Web Scraping

Before scraping, we need some Python libraries.

1 Install `requests`

Used to **fetch webpage content**

```
pip install requests
```

2 Install `BeautifulSoup`

Used to **parse and extract HTML data**

```
pip install beautifulsoup4
```

3 Install `pandas`

Used to **store and analyze scraped data**

```
pip install pandas
```



What Each Library Does

Library	Purpose
<code>requests</code>	Sends HTTP requests to websites
<code>BeautifulSoup</code>	Extracts data from HTML
<code>pandas</code>	Saves data as tables (CSV, Excel)



Summary

- Web scraping = automatic data collection from websites
- Used in data analysis, AI, business, and research
- Ethics and legality **must be respected**
- Static sites are easier than dynamic ones
- Python libraries make scraping simple and powerful